

Maurizio Valle

professor

基本信息

姓名: Maurizio Valle

所属院校: 热那亚大学

地区: Italy

学科领域

人工智能

电子与计算机工程

机器人工程

生物医学工程

智能装备与系统

集成电路设计与集成系统

研究领域

深度学习与神经网络

智能感知与传感器技术

嵌入式系统与硬件设计

机器学习理论

机器人控制与运动规划

计算机视觉与感知系统

康复工程与辅具设计

FPGA与可重构计算

智能传感与工业物联网

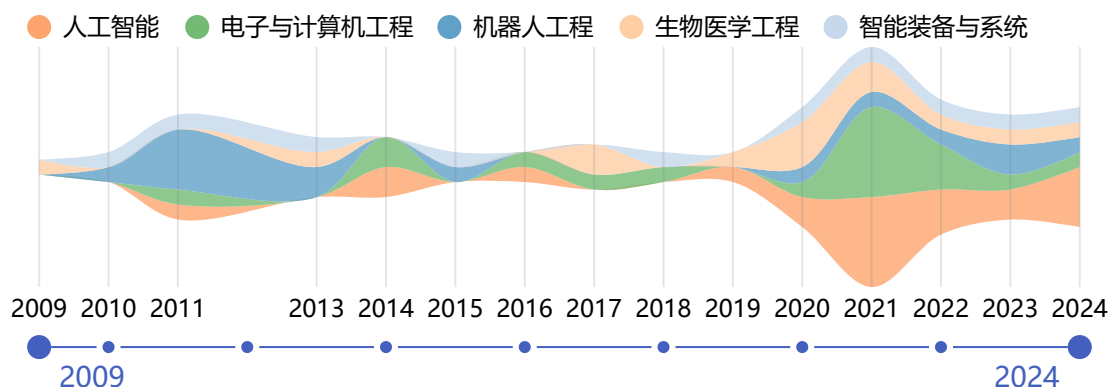
生物信号处理与模式识别

+2

个人简介

Maurizio Valle is an ordinary professor at the Department of Naval, Electrical, Electronic and Telecommunications Engineering - DITEN. He teaches IINF - 01/A – Electronics. He serves as the coordinator of the doctoral research at DITEN. His contact numbers are (+39) 010 33 52775 and (+39) 3483053884. Information about his office location is restricted to UniGePASS.

近年发表的论文



On Edge Human Action Recognition Using Radar-Based Sensing and Deep Learning

IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS

Gianoglio, Christian; Mohanna, Ammar; Rizik, Ali; Moroney, Laurence; Valle, Maurizio

被引用次数: 1

2024

Neuromorphic Tactile Sensing System for Textural Features Classification

IEEE SENSORS JOURNAL

Ali, Haydar Al Haj; Abbass, Yahya; Gianoglio, Christian; Ibrahim, Ali; Oh, Changjae; Valle, Maurizio

被引用次数: 1

2024

Texture Perception Using Tactile Sensing Glove Based on PVDF Sensors and Machine Learning

IEEE SENSORS LETTERS

Abbass, Yahya; Gianoglio, Christian; Ali, Haydar Al Haj; Saleh, Moustafa; Valle, Maurizio

被引用次数: 0

2024

Hardness Discrimination Using Piezoelectric-Based Biomimetic Tactile Sensor and Machine Learning

IEEE SENSORS LETTERS

Bassal, Hussein; Abbass, Yahya; Gianoglio, Christian; Valle, Maurizio

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2024

Trade-off Between Accuracy and Computational Cost With Neural Architecture Search: A Novel Strategy for Tactile Sensing Design

IEEE SENSORS LETTERS

Gianoglio, Christian; Ragusa, Edoardo; Gastaldo, Paolo; Valle, Maurizio

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2023

FPGA-Based Tactile Sensory Platform with Optical Fiber Data Link for Feedback Systems in Prosthetics

ELECTRONICS

Stanchieri, Guido Di Patrizio; Saleh, Moustafa; De Marcellis, Andrea; Ibrahim, Ali; Faccio, Marco; Valle, Maurizio; Palange, Elia

被引用次数: 2

2023

Pilot Study: Response of a Piezoelectric Polymer based Sensing system to Indentation

2023 IEEE SENSORS

Yaacoub, Mohamad; Abbass, Yahya; Gianoglio, Christian; Seminara, Lucia; Valle, Maurizio

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2023

Workspace Scaling in Virtual Reality based Robot Teleoperation

PROCEEDINGS OF THE 4TH AUGMENTED HUMANS INTERNATIONAL CONFERENCE 2023, AHS2023

Omarali, Bukeikhan; Javaid, Shafiq; Valle, Maurizio; Farkhatdinov, Ildar

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2023

Embedded real-time objects' hardness classification for robotic grippers

FUTURE GENERATION COMPUTER SYSTEMS-THE INTERNATIONAL JOURNAL OF ESCIENCE

Amin, Youssef; Gianoglio, Christian; Valle, Maurizio

被引用次数: 6

2023

Spiking Neural Network Based on Threshold Encoding For Texture Recognition

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Ali, Haydar Al Haj; Dabbous, Ali; Ibrahim, Ali; Valle, Maurizio

被引用次数: 1

2022

Object Contact Shape Classification Using Neuromorphic Spiking Neural Network with STDP Learning

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Dabbous, Ali; Ibrahim, Ali; Alameh, Mohamad; Valle, Maurizio; Bartolozzi, Chiara

被引用次数: 4

2022

A Pipelined Implementation of the n-mode Tensor-Matrix Multiplication

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Ragusa, Edoardo; Gianoglio, Christian; Zunino, Rodolfo; Valle, Maurizio; Gastaldo, Paolo

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2022

A Convolutional Neural Network-Based Method for Discriminating Shadowed Targets in Frequency-Modulated Continuous-Wave Radar Systems

SENSORS

Mohanna, Ammar; Gianoglio, Christian; Rizik, Ali; Valle, Maurizio

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2022

A Novel Learning Strategy for the Trade-Off Between Accuracy and Computational Cost: A Touch Modalities Classification Case Study

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Gianoglio, Christian; Ragusa, Edoardo; Gastaldo, Paolo; Valle, Maurizio

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2022

Full-hand electrotactile feedback using electronic skin and matrix electrodes for high-bandwidth human-machine interfacing

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND
ENGINEERING SCIENCES

Abbass, Yahya; Dosen, Strahinja; Seminara, Lucia; Valle, Maurizio

被引用次数: 8

2022

Approximate Computing Circuits for Embedded Tactile Data Processing

ELECTRONICS

Osta, Mario; Ibrahim, Ali; Valle, Maurizio

被引用次数: 0

2022

Novel Wearable Tactile Feedback System for poststroke Rehabilitation

2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)

Abbass, Yahya; Seminara, Lucia; Saleh, Moustafa; Valle, Maurizio

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2021

Touch Modality Classification using Spiking Neural Networks and Supervised-STDP Learning

2021 28TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS, AND SYSTEMS (IEEE ICECS 2021)

Dabbous, Ali; Ibrahim, Ali; Valle, Maurizio; Bartolozzi, Chiara

被引用次数: 0

2021

Hybrid Fixed-point/Binary Convolutional Neural Network Accelerator for Real-time Tactile Processing

2021 28TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS, AND SYSTEMS (IEEE ICECS 2021)

Younes, Hamoud; Ibrahim, Ali; Rizk, Mostafa; Valle, Maurizio

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2021

Workspace Scaling and Rate Mode Control for Virtual Reality based Robot Teleoperation

2021 IEEE INTERNATIONAL CONFERENCE ON SYSTEMS, MAN, AND CYBERNETICS (SMC)

Omarali, Bukeikhan; Althoefer, Kaspar; Mastrogiovanni, Fulvio; Valle, Maurizio; Farkhatdinov, Ildar

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2021

Touch Modality Classification Using Recurrent Neural Networks

IEEE SENSORS JOURNAL

Alameh, Mohamad; Abbass, Yahya; Ibrahim, Ali; Moser, Gabriele; Valle, Maurizio

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2021

Efficient FPGA Implementation of Approximate Singular Value Decomposition based on Shallow Neural Networks

2021 IEEE 3RD INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS (AICAS)

Younes, Hamoud; Ibrahim, Ali; Rizk, Mostafa; Valle, Maurizio

被引用次数: 4

2021

Embedded Electrotactile Feedback System for Hand Prostheses Using Matrix Electrode and Electronic Skin

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Abbass, Yahya; Saleh, Moustafa; Dosen, Strahinja; Valle, Maurizio

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2021

Algorithmic-Level Approximate Tensorial SVM Using High-Level Synthesis on FPGA

ELECTRONICS

Younes, Hamoud; Ibrahim, Ali; Rizk, Mostafa; Valle, Maurizio

被引用次数: 14

2021

A Shallow Neural Network for Real-Time Embedded Machine Learning for Tensorial Tactile Data Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Younes, Hamoud; Ibrahim, Ali; Rizk, Mostafa; Valle, Maurizio

被引用次数: 2

2021

Validation of Screen-Printed Electronic Skin Based on Piezoelectric Polymer Sensors

SENSORS

Fares, Hoda; Abbass, Yahya; Valle, Maurizio; Seminara, Lucia

被引用次数: 17

2020

Editorial of Special Issue Tactile Sensing Technology and Systems

MICROMACHINES

Valle, Maurizio

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2020

Smart Tactile Sensing Systems Based on Embedded CNN Implementations

MICROMACHINES

Alameh, Mohamad; Abbass, Yahya; Ibrahim, Ali; Valle, Maurizio

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2020

Introduction to the Special Issue on the 2nd IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS 2020)

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Serrano-Gotarredona, Teresa; Valle, Maurizio; Conti, Francesco; Li, Hai

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2020

Dual-Parameter Modulation Improves Stimulus Localization in Multichannel Electrotactile Stimulation

IEEE TRANSACTIONS ON HAPTICS

Seminara, Lucia; Fares, Hoda; Franceschi, Marta; Valle, Maurizio; Strbac, Matija; Farina, Dario; Dosen, Strahinja

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2020

Temporal Asynchrony but Not Total Energy Nor Duration Improves the Judgment of Numerosity in Electrotactile Stimulation

FRONTIERS IN BIOENGINEERING AND BIOTECHNOLOGY

Nataletti, Sara; Leo, Fabrizio; Seminara, Lucia; Trompetto, Carlo; Valle, Maurizio; Dosen, Strahinja; Brayda, Luca

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2020

Experimental Assessment of the Interface Electronic System for PVDF-Based Piezoelectric Tactile Sensors

SENSORS

Saleh, Moustafa; Abbass, Yahya; Ibrahim, Ali; Valle, Maurizio

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Chang, Robert Chen-Hao; Lee, Gwo Giun (Chris); Delbruck, Tobj; Valle, Maurizio

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2019

CMOS event-driven tactile sensor circuit

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Abou Khalil, Ali; Valle, Maurizio; Chible, Hussein; Bartolozzi, Chiara

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2018

Experimental characterization of dedicated front-end electronics for piezoelectric tactile sensing arrays

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Ibrahim, Ali; Pinna, Luigi; Valle, Maurizio

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Real-Time Embedded Machine Learning for Tensorial Tactile Data Processing

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Time-based calibration-less read-out circuit for interfacing wide range MOX gas sensors

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Hijazi, Zeinab; Grassi, Marco; Caviglia, Daniele D.; Valle, Maurizio

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2018

A System for Electrotactile Feedback Using Electronic Skin and Flexible Matrix Electrodes: Experimental Evaluation

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Franceschi, Marta; Seminara, Lucia; Dosen, Strahinja; Strbac, Matija; Valle, Maurizio; Farina, Dario

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2017

Spike-Based Readout of POSFET Tactile Sensors

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Caviglia, Stefano; Pinna, Luigi; Valle, Maurizio; Bartolozzi, Chiara

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2017

Interface Electronics for Tactile Sensors Based on Piezoelectric Polymers

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Pinna, Luigi; Ibrahim, Ali; Valle, Maurizio

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2017

Real-Time Digital Signal Processing Based on FPGAs for Electronic Skin Implementation SENSORS

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Model-Based Simulation Framework for FlexRay Communication Systems

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Muller, Candice; Valle, Maurizio; Buzas, Roman; Brtan, Filip

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2016

Towards integrating intelligence in electronic skin

MECHATRONICS

Seminara, Lucia; Pinna, Luigi; Ibrahim, Ali; Noli, Luca; Caviglia, Stefano; Gastaldo, Paolo; Valle, Maurizio

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2016

Tactile data processing method for the reconstruction of contact force distributions

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Directions toward effective utilization of tactile skin: A review Reply (vol 14, 2014)

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2014

Tactile Sensing Chips With POSFET Array and Integrated Interface Electronics

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Dahiya, Ravinder S.; Adami, Andrea; Pinna, Luigi; Collini, Cristian; Valle, Maurizio; Lorenzelli, Leandro

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A Tensor-Based Pattern-Recognition Framework for the Interpretation of Touch Modality in Artificial Skin Systems

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Gastaldo, Paolo; Pinna, Luigi; Seminara, Lucia; Valle, Maurizio; Zunino, Rodolfo

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Computational Intelligence Techniques for Tactile Sensing Systems

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Piezoelectric Polymer Transducer Arrays for Flexible Tactile Sensors

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Seminara, Lucia; Pinna, Luigi; Valle, Maurizio; Basirico, Laura; Loi, Alberto; Cosseddu, Piero; Bonfiglio, Annalisa; Ascia, Alberto; Biso, Maurizio; Ansaldo, Alberto; Ricci, Davide; Metta, Giorgio

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POSFET touch sensing transducers: Interface electronics design methodology based on the transconductance-to-drain-current efficiency gm/ID

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Barboni, Leonardo; Valle, Maurizio

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CHARGE AMPLIFIER DESIGN METHODOLOGY FOR PVDF-BASED TACTILE SENSORS

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Towards Tactile Sensing System on Chip for Robotic Applications

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2011

Electromechanical characterization of piezoelectric PVDF polymer films for tactile sensors in robotics applications

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Seminara, Lucia; Capurro, Marco; Cirillo, Paolo; Cannata, Giorgio; Valle, Maurizio

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2011

Tactile-Data Classification of Contact Materials Using Computational Intelligence

IEEE TRANSACTIONS ON ROBOTICS

Decherchi, Sergio; Gastaldo, Paolo; Dahiya, Ravinder S.; Valle, Maurizio; Zunino, Rodolfo

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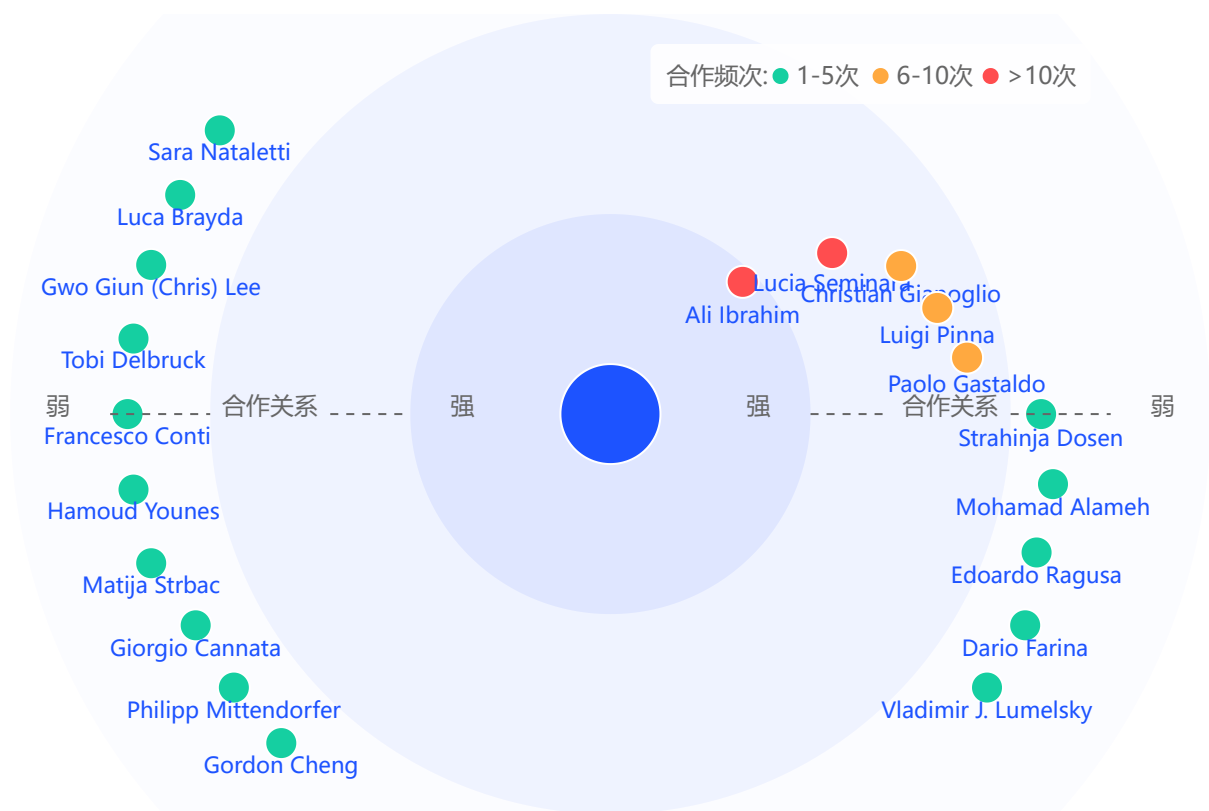
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Tactile Sensing-From Humans to Humanoids	
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SPICE Model for Lossy Piezoelectric Polymers (vol 56, pg 387, 2009)	
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Dahiya, Ravinder S.; Valle, Maurizio; Lorenzelli, Leandro	
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SPICE Model for Lossy Piezoelectric Polymers	
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Dahiya, Ravinder S.; Valle, Maurizio; Lorenzelli, Leandro	
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Low-noise low-power CMOS preamplifier for multisite extracellular neuronal recordings	
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Bottino, Emanuele; Massobrio, Paolo; Martinoia, Sergio; Pruzzo, Giacomo; Valle, Maurizio	
被引用次数: 8	2009
High abstraction level CAD tool implementation of MOS drain current models	
MICROELECTRONICS JOURNAL	
Prodanov, William; Valle, Maurizio	
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Piezoelectric oxide semiconductor field effect transistor touch sensing devices	
APPLIED PHYSICS LETTERS	
Dahiya, Ravinder S.; Metta, Giorgio; Valle, Maurizio; Adami, Andrea; Lorenzelli, Leandro	
被引用次数: 144	2009
A COMPARATIVE STUDY OF VARIOUS PROBABILITY DENSITY ESTIMATION METHODS FOR DATA ANALYSIS	
INTERNATIONAL JOURNAL OF COMPUTATIONAL INTELLIGENCE SYSTEMS	
Assenza, Alex; Valle, Maurizio; Verleysen, Michel	
被引用次数: 6	2008

学术合作



Ali Ibrahim

合作次数: 18

热那亚大学

Lucia Seminara

合作次数: 13

热那亚大学

Christian Gianoglio

合作次数: 10

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Luigi Pinna

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